PK FOKAM RESEARCH CENTER

MICROFI PROJECT — SPECIFICATIONS

Textual Use Case Document — Subsystems 1 & 2

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Abstract

This document presents the official, fully articulated textual specifications for all 13 Use Cases of the **MICROFI** System (Microfinance Collection and Management System), developed for PK FOKAM RESEARCH CENTER.

Contents

1 Subsystem 1: Mobile App (TPU)

1.1 UC01 – Authenticate

Identifier	**UC01**
Name	Authenticate
Actor(s)	Collection Agent
Objective	Authenticate the Collection Agent on the mobile terminal to grant access to field collection features.
Pre-conditions	Terminal powered on, MICROFI application launched, active agent account.
Post-conditions	Agent authenticated, local session established, access token generated.
Nominal Scenario	1. The agent reaches the login screen of the mobile application. 2. The system displays the credentials form (User ID / PIN code). 3. The agent enters credentials. 4. The system validates credentials (locally offline or against the server if online). 5. The system loads the agent's profile and parameters. 6. The system displays the main collection dashboard.
Alternative Scenarios	**4a: Invalid credentials**: System displays "Invalid User ID or PIN", attempt counter +1, returns to step 3. **4b: Locked account**: System displays "Account locked, contact your manager".
Business Rules	• Account locked after 3 consecutive failed attempts. • Confidential PIN code (4 to 6 digits). • Offline authentication backed by an encrypted local key.

1.2 UC02 – Register Client Deposit

Identifier	**UC02**
Name	Register Client Deposit
Actor(s)	Collection Agent
Objective	Record a cash deposit made by a client to the collection agent on the field.
Pre-conditions	Agent authenticated, collection session active.
Post-conditions	Transaction saved in local database, physical receipt printed, synchronization initiated if network is available.
Nominal Scenario	1. The agent navigates to ”New Deposit”. 2. The system displays the transaction form. 3. The agent inputs the client's account number and deposit amount. 4. The system automatically includes limit checking (**UC03: Check Transaction Limit**). 5. The agent validates the transaction. 6. The system saves the transaction into local terminal storage. 7. The system automatically includes receipt printing (**UC04: Print Deposit Receipt**). 8. The system displays ”Deposit registered successfully”.
Alternative Scenarios	**4a: Limit exceeded**: System displays an error message and blocks validation. **8a: Network available**: System triggers **UC05 (Manual Synchronisation)** to push pending transactions to the backend.
Business Rules	• Client account identifier and amount are mandatory. • Automatic timestamping and GPS tagging of each transaction. • Validated transactions cannot be modified or deleted locally.

1.3 UC03 – Check Transaction Limit

Identifier	**UC03**
Name	Check Transaction Limit
Actor(s)	System / Mobile App (Included by *Register Client Deposit*)
Objective	Verify that the transaction amount complies with individual transaction limits and the agent's daily threshold.
Pre-conditions	Deposit amount entered by the agent.
Post-conditions	Transaction authorized or rejected.
Nominal Scenario	1. System retrieves the entered deposit amount. 2. System reads the agent's configured limit parameters. 3. System computes the total accumulated deposits for the current day. 4. System checks if $Amount \leq Single\_Limit$ AND $Daily\_Total + Amount \leq Daily\_Limit$. 5. System validates controls and allows deposit processing.
Alternative Scenarios	**4a: Single limit exceeded**: System returns refusal "Amount exceeds single transaction limit". **4b: Daily limit exceeded**: System returns refusal "Agent daily limit reached".
Business Rules	• Limits are set by the Branch Manager during session opening. • Enforcement performed locally without requiring network connection.

1.4 UC04 – Print Deposit Receipt

Identifier	**UC04**
Name	Print Deposit Receipt
Actor(s)	System / Thermal Printer (Included by *Register Client Deposit*)
Objective	Issue a physical printed receipt to give to the client following a validated deposit.
Pre-conditions	Deposit successfully validated and saved locally.
Post-conditions	Physical receipt printed.
Nominal Scenario	1. System generates receipt layout (Date, Time, Agent ID, Masked Account No, Amount, Encrypted Tx ID). 2. System sends print stream to integrated Bluetooth/Thermal printer. 3. Printer outputs the paper receipt. 4. System confirms successful printing.
Alternative Scenarios	**3a: Out of paper / Printer offline**: System displays "Printing error", enables "Reprint receipt" once printer is ready.
Business Rules	• Systematic printing for every validated deposit. • Printed receipt must include a unique encrypted security hash.

1.5 UC05 – Manual Synchronisation

Identifier	UC05
Name	Manual Synchronisation
Actor(s)	Collection Agent
Objective	Transmit locally stored offline transactions to the Back-Office database.
Pre-conditions	Agent authenticated, offline transactions pending, network connection available.
Post-conditions	Transactions uploaded, validated by server, and marked as ”Synchronized” on the terminal.
Nominal Scenario	1. The agent clicks ”Synchronize”. 2. System checks network connectivity. 3. System bundles un-synchronized local transactions into an encrypted payload. 4. System sends payload to the Back-Office Middleware. 5. System receives server acknowledgment (ACK). 6. System updates local transaction statuses and displays ”Synchronization successful”.
Alternative Scenarios	**2a: No network**: System displays ”Network unavailable, try again later”. **5a: Transmission failure**: System resends unconfirmed batches.
Business Rules	• Data retained locally until server acknowledgment is confirmed. • Encrypted batch processing.

2 Subsystem 2: Back-Office

2.1 UC06 – Connect

Identifier	**UC06**
Name	Connect
Actor(s)	User (*Branch Manager, Cashier, System Administrator*)
Objective	Authenticate Back-Office users to grant role-based access to management interfaces.
Pre-conditions	User account active in Back-Office system.
Post-conditions	User authenticated, active web session established.
Nominal Scenario	1. User navigates to the Back-Office portal. 2. System presents authentication form. 3. User enters username and password. 4. System validates credentials and role permissions. 5. System redirects user to their designated dashboard based on role.
Alternative Scenarios	**4a: Invalid credentials**: System displays ”Invalid username or password”.
Business Rules	• Single Sign-On (SSO) compatible. • Session timeout after 15 minutes of inactivity.

2.2 UC07 – Affect Terminal to Agent

Identifier	**UC07**
Name	Affect Terminal to Agent
Actor(s)	Branch Manager
Objective	Assign a hardware TPU terminal to a specific Collection Agent for field operations.
Pre-conditions	Branch Manager connected, Terminal created (UC14), Agent account active (UC13a).
Post-conditions	Terminal bound to the specified Collection Agent.
Nominal Scenario	1. Manager selects ”Terminal Management”. 2. System lists available terminals and active agents. 3. Manager selects a terminal IMEI/Serial and assigns it to an agent. 4. System updates binding records in the database. 5. System displays ”Terminal successfully assigned”.
Alternative Scenarios	**3a: Terminal already assigned**: System prompts ”Reassign terminal to new agent?”.
Business Rules	• One active agent per terminal at any given time.

2.3 UC08 – Geolocalise Agent

Identifier	**UC08**
Name	Geolocalise Agent
Actor(s)	Branch Manager, Mobile APP MICROFI (*Secondary System Actor*)
Objective	Track agent geographic positions and field routes on the Back-Office map dashboard.
Pre-conditions	Branch Manager connected, GPS coordinates emitted by Mobile APP MICROFI.
Post-conditions	Real-time coordinates and historical routes displayed on map.
Nominal Scenario	1. Branch Manager accesses ”Field Supervision Map”. 2. System fetches latest coordinates provided by Mobile APP MICROFI. 3. System renders agent positions on interactive map interface. 4. Manager selects an agent marker to view session details (last deposit time, battery level).
Alternative Scenarios	**2a: GPS signal lost**: System displays last known position with alert ”Signal lost”.
Business Rules	• Coordinates refreshed periodically (every 5–10 minutes).

2.4 UC09 – Open Agent Session

Identifier	**UC09**
Name	Open Agent Session
Actor(s)	Cashier, Branch Manager
Objective	Open an agent’s daily collection session, assigning operational limits and initial floats.
Pre-conditions	Cashier/Manager connected to Back-Office.
Post-conditions	Agent session active and authorized for field collection.
Nominal Scenario	1. Cashier opens ”Agent Session Management”. 2. System lists branch collection agents. 3. Cashier selects an agent and clicks ”Open Session”. 4. Cashier defines session parameters (daily limit, starting balance). 5. Cashier validates session opening. 6. System stores session state and generates security tokens for terminal sync.
Alternative Scenarios	**3a: Session already open**: System warns ”Agent already has an active session”. **4a: Extension Choose Transaction**: Cashier selects pre-assigned transaction templates via **UC09a**.
Business Rules	• Only one active collection session per agent at a time.

UC09a – Choose Transaction (Extends UC09)

Optionally allows Cashier/Manager to select specific transaction types or pre-allocated client lists assigned to an agent's session during setup.

2.5 UC10 – Transaction Validation

Identifier	UC10
Name	Transaction Validation
Actor(s)	Cashier, CBS (Core Banking System)
Objective	Validate collected funds, reconcile physical cash with digital logs, and post entries to the CBS.
Pre-conditions	Agent returned from field, session synchronized.
Post-conditions	Session closed, variances computed, accounting entries posted to CBS.
Nominal Scenario	1. Cashier selects an agent's session to close. 2. System includes physical cash counting (**UC10b: Bill Breakdown**). 3. System includes automatic reconciliation (**UC10a: Calculate Cash-Digital Difference**). 4. Cashier confirms session closure. 5. System transmits validated entries to the **CBS** for posting. 6. **CBS** returns confirmation acknowledgment. 7. System updates session status to "Closed & Validated".
Alternative Scenarios	**3a: Variance detected**: System executes extension **UC10c (Flag Suspicious Transaction).** **5a: CBS timeout**: System queues entries and notifies Cashier.
Business Rules	• Bill breakdown mandatory for session closure. • Unresolved variances exceeding tolerance threshold must be flagged.

Sub-components for UC10:

- **UC10a – Calculate Cash-Digital Difference (Included):** Computes variance: $Variance = PhysicalCash - DigitalTotal$.
- **UC10b – Bill Breakdown (Included):** Captures physical banknote and coin counts by denomination (10 000, 5 000, 2 000 FCFA, etc.).
- **UC10c – Flag Suspicious Transaction (Extends):** Marks session/transaction for audit if variance exceeds threshold.

2.6 UC11 – Consult Account

Identifier	UC11
Name	Consult Account
Actor(s)	System Administrator, Branch Manager, Cashier
Objective	View client account balances, transaction histories, and account statuses within the Back-Office.
Pre-conditions	Authorized Back-Office user connected.
Post-conditions	Client account details displayed.
Nominal Scenario	1. User navigates to ”Account Inquiry”. 2. User enters account number or client name. 3. System fetches record from database. 4. System displays balance, recent collection history, and status.
Alternative Scenarios	**3a: Account not found**: System displays ”No account found matching search criteria”.
Business Rules	• Restricted to Back-Office users to guarantee data privacy on field terminals.

2.7 UC12 – Consult Flagged Transactions

Identifier	UC12
Name	Consult Flagged Transactions
Actor(s)	System Administrator, Branch Manager
Objective	Review transactions or sessions flagged for anomalies, cash shortages, or compliance audits.
Pre-conditions	Administrator or Manager connected.
Post-conditions	Flagged transaction reports reviewed and audit decisions logged.
Nominal Scenario	1. User opens ”Audit & Flagged Transactions”. 2. System displays list of flagged items with variance reasons. 3. User selects an entry to inspect detailed logs (GPS, receipts, bill breakdown). 4. User records audit decision (Approve adjustment, Reject, Escalated).
Alternative Scenarios	**2a: No flagged items**: System displays ”No pending flagged transactions”.
Business Rules	• Audit decisions logged permanently for compliance tracing.

2.8 UC13 – Create Account (Generic Case)

Identifier	UC13
Name	Create Account (Generic Case)
Actor(s)	System Administrator
Objective	Provision new user accounts in the Back-Office platform.
Note	Generic Use Case. Specialized by UC13a, UC13b, and UC13c.
Pre-conditions	System Administrator connected.
Post-conditions	User account created and credentials generated.
Nominal Scenario	1. Administrator accesses ”User Management”. 2. System presents user creation form. 3. Administrator inputs primary user info (Name, Email, Phone). 4. System verifies uniqueness of username/email. 5. System creates base account and generates temporary password.
Alternative Scenarios	**4a: Duplicate user**: System displays ”User already exists”, returns to step 3.
Business Rules	• Mandatory password change upon first login. • Email/Username must be globally unique.

Specializations of UC13:

- **UC13a – Create Agent Account:** Assigns unique Agent Code, branch binding, default limits, and mobile PIN.
- **UC13b – Create Cashier Account:** Assigns cashier desk/vault, validation rights, and CBS Cashier ID.
- **UC13c – Create Branch Manager Account:** Grants branch supervision, tracking, and audit access.

2.9 UC14 – Create Terminal

Identifier	**UC14**
Name	Create Terminal
Actor(s)	System Administrator
Objective	Register new TPU hardware terminals into the system inventory.
Pre-conditions	System Administrator connected.
Post-conditions	Terminal hardware registered and ready for agent allocation (UC07).
Nominal Scenario	1. Administrator accesses ”Terminal Inventory”. 2. System displays terminal registration form. 3. Administrator enters Serial Number, IMEI, model, and MAC address. 4. System validates uniqueness of Serial/IMEI. 5. System saves terminal into active inventory.
Alternative Scenarios	**4a: Duplicate IMEI/Serial**: System displays ”Terminal hardware already registered”.
Business Rules	• Unique Serial Number and IMEI per hardware unit.